The Math Behind `text-diffusion-fashion-mnist`

A from-scratch text-conditioned DDPM

This note collects the equations implemented in the repository, mapping each to the file where it lives. The model is a miniature Stable Diffusion: a U-Net learns to reverse a Gaussian noising process, conditioned on frozen CLIP text embeddings and steered at sample time by classifier-free guidance.

1 Forward process (adding noise)

src/diffusion.py — `q_sample`

Given a clean image $\mathbf{x}_0$, the forward (diffusion) process gradually corrupts it over T timesteps with Gaussian noise governed by a variance schedule $\beta_1, \ldots, \beta_T$. Define $\alpha_t = 1 - \beta_t$ and the cumulative product $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. A key property is that we can sample $\mathbf{x}_t$ at *any* timestep in closed form, without iterating:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\big(\mathbf{x}_t;\ \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\ (1-\bar{\alpha}_t)\,\mathbf{I}\big), \tag{1}$$

which, via the reparameterization trick, is implemented as

$$\boxed{\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).} \tag{2}$$

2 Cosine noise schedule

src/diffusion.py — `make_beta_schedule`

Rather than a linear β_t, we use the cosine schedule of Nichol & Dhariwal, which destroys information more gradually and trains better at low resolution:

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \qquad f(t) = \cos\left(\frac{t/T + s}{1+s} \cdot \frac{\pi}{2}\right)^2, \qquad s = 0.008, \tag{3}$$

and the per-step variances are recovered (and clipped for stability) as

$$\beta_t = \min\left(1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}},\ 0.999\right). \tag{4}$$

3 Training objective

src/train.py — training loop

The U-Net $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, c)$ is trained to predict the noise $\boldsymbol{\epsilon}$ that was added, given the timestep t and the text conditioning c. This yields the simple denoising objective

$$\boxed{\mathcal{L}_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0,\ \boldsymbol{\epsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I}),\ t\sim\mathcal{U}\{1,T\}}\Big[\ \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, c)\|^2\ \Big].} \tag{5}$$

During training the caption c is replaced by a null token $\varnothing$ with probability p_{drop} (here 0.15), so the network also learns the unconditional score $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \varnothing)$.

4 Text conditioning (the multimodal bridge)

src/text_encoder.py, src/unet.py

Each Fashion-MNIST class is turned into a caption and encoded by a *frozen* CLIP text encoder into token embeddings $c \in \mathbb{R}^{L \times d}$. The U-Net consumes these two ways: a pooled vector is added to the timestep embedding (FiLM), and cross-attention lets image positions (queries) attend to the text tokens (keys/values):

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \qquad Q = W_Q\, h_{\mathrm{img}}, \quad K = W_K\, c, \quad V = W_V\, c. \tag{6}$$

5 Classifier-free guidance (sampling)

src/diffusion.py — `ddim_sample`

At sample time we combine the conditional and unconditional noise predictions and extrapolate, with guidance scale s trading diversity for prompt adherence:

$$\boxed{\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t, c) = \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \varnothing) + s\left(\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, c) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \varnothing)\right).} \tag{7}$$

6 DDIM sampling (the reverse process)

src/diffusion.py — `ddim_sample`

Instead of T stochastic steps we take a short subsequence of timesteps and apply the (deterministic when $\eta = 0$) DDIM update. First predict the clean image from the guided noise,

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\, \tilde{\boldsymbol{\epsilon}}_\theta}{\sqrt{\bar{\alpha}_t}}, \tag{8}$$

then step to the previous timestep with noise scale

$$\sigma_t = \eta \sqrt{\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}} \sqrt{1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}}, \tag{9}$$

$$\boxed{\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\, \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\, \tilde{\boldsymbol{\epsilon}}_\theta + \sigma_t\, \mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).} \tag{10}$$

With $\eta = 0$ the $\sigma_t \mathbf{z}$ term vanishes and sampling is fully deterministic.

References. Ho et al., *DDPM* (2020); Nichol & Dhariwal, *Improved DDPM* (2021); Song et al., *DDIM* (2021); Ho & Salimans, *Classifier-Free Guidance* (2022); Rombach et al., *Latent Diffusion / Stable Diffusion* (2022).